

ServiceNow Scheduled Jobs

Scheduled Jobs Overview

Scheduled Jobs are server-side automations that execute tasks automatically at a specific time or recurring interval without manual intervention.

Purpose: Used to automate routine and administrative tasks in ServiceNow.

Examples

- Send daily reports
- Clean up old records
- Update data automatically
- Run scheduled imports
- Generate notifications

We can automate the following tasks below

Scheduled Jobs can be used to automate the following tasks:

- Automatically create records such as:
 - Incidents
 - Problems
 - Changes
 - Configuration Items (CIs)
 - Other records from templates
- Automatically run and distribute reports.
- Execute Business Rules or custom scripts on a schedule.
- Schedule tasks to run:
 - Daily
 - Weekly

- Monthly (e.g., end of month)
 - Weekdays only
- Execute scheduled jobs from scripts when required.
- Trigger email notifications automatically.

Some scenarios where we can use schedule jobs in ServiceNow

- Deliver reports automatically to stakeholders:
 - Daily
 - Weekly
 - Monthly
 - Custom schedules
- Create recurring tasks automatically and assign them to teams.
- Send reports with complete details through email.
- Auto-close Incident tickets that are in **Pending Customer Action** when the customer does not respond for a specified period.
- Auto-close Incident tickets that remain in **Resolved** state for a defined period (e.g., 9 months).
- Trigger events, scripts, notifications, or maintenance activities on a weekly, monthly, or custom schedule.

Scheduled Jobs Tables

Type	Purpose	Example
sysauto_script	Run scripts	Auto-close tickets
scheduled_import_set	Import data	HR data sync
sysauto_template	Create records	Monthly maintenance task
sysauto_report	Send reports	Weekly incident report

Scheduled job states

A scheduled job can be in any one of the following states.

Ready: Waiting for scheduled time
Job is ready to run at the next scheduled interval

Running: Currently executing
Job is in the process of carrying out a task

Queued: Waiting in scheduler queue
Job has been added to the scheduler queue and is waiting to run

Error: Execution failed
Error occurred while running the job

Create a new scheduled job

Automate generation and distribution of a report

Exercise: 1

Generate and distribute scheduled reports via email.

Procedure

1. Navigate to **System Definition > Scheduled Jobs.**
2. **Click New.**
3. **Select** and **click on** Automate the generation and distribution of a report
4. Fulfil scheduled jobs form
5. **Name:** Category Of Incidents
6. **Report:** Category of Incidents
7. **Users:** srinivassnow2113@gmail.com
8. **Email addresses:** srinivassnow2113@gmail.com
9. **Active:** Checked
10. **Run:** Monthly
11. **Day:** 15
12. **Time:** Hours 04, 34, 00
13. **Type:** PDF-landscape
14. Click on **Submit**

Before testing this, we should set up particular time zone for this instance

1. Navigate to **Setting→Time Zone**
2. Select **GMT** time Zone

Create a new scheduled Job

All resolved incidents should get closed automatically on **every 6 months**

Exercise: 2

All resolved incidents should get closed automatically on **every 6 months**

Procedure

1. Navigate to **System Definition > Scheduled Jobs**.
2. **Click New**.
3. **Select** and **click on** Automatically run a script of your choosing
4. Fulfil scheduled jobs form
5. **Name:** All resolved incidents should get closed automatically
6. **Active:** True
7. **Run:** Weekly
8. **Day:** Friday
9. **Time Zone:** GMT
10. **Time:** 16:20:00
11. Click on **Submit**

Run this Script

```
var eq = 'resolved_atONLast 6  
months@javascript:gs.beginningOfLast6Months()@javascript:gs.endOf  
Last6Months()';
```

```
var gr = new GlideRecord('incident');  
gr.addEncodedQuery(eq);  
gr.query();  
while(gr.next()){
```

```
gr.state = '7';  
gr.work_notes = 'All resolved incidents are closing through scheduled  
jobs';  
gr.update();  
}
```

Create a new scheduled Job

Create an incident ticket every week on Saturday in the mng 9AM

Exercise: 3

Create an incident ticket every week on Saturday in the mng 9AM

Procedure

1. Navigate to **System Definition > Scheduled Jobs.**
2. **Click New.**
3. **Select** and **click on** Automatically generate something (a change, an incident, a ci, etc) from a template
4. Create an incident ticket every week on Saturday in the mng 9AM
5. **Active:** True
6. **Run:** Weekly
7. **Time Zone:** GMT
8. **Time Hours:** 14:20:00
9. **Generate This:** Network Servers Maintenance
10. Create **Template**

Scheduled Entity Generation form fields:

Name: Create an incident ticket every week on Saturday in the mng 9AM

Active: Checked

Run: Weekly

Day: Saturday

Time zone: GMT

Time: Hours 14, 20, 00

Generate this: Network Servers Maintanance

Template Form Fields

1. **Name:** Network Servers maintenance
2. **Table:** Incident
3. **Active:** True
4. **Short description:** Network Server Maintanance
5. **Category:** Network
6. **Subcategory:** DNS
7. **Caller:** ITIL User